



BITS Pilani

Pilani | Dubai | Goa | Hyderabad

Introduction to Python for Data Science

DSECLPFDS / AIMLCPFDS

- Exception Handling
- Functions
- Files

Agenda for CS#4



- SciPy Ecosystem
- Basics of Numpy
- Basics of Pandas
- Visualizations using Matplotlib
- Visualizations using Seaborn

Introduction to SciPy Ecosystem

SciPy (pronounced “Sigh Pie”) is a Python-based ecosystem of open-source software for mathematics, science, and engineering. In particular, these are some of the core packages:



NumPy
Base N-dimensional
array package



SciPy library
Fundamental library for
scientific computing



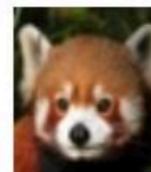
Matplotlib
Comprehensive 2-D
plotting



IPython
Enhanced interactive
console



SymPy
Symbolic mathematics



pandas
Data structures &
analysis

<https://www.scipy.org/>
<https://www.scipy.org/about.html>

Numpy

NumPy stands for Numerical Python.

- NumPy is a Python library used for working with arrays.
- It also has functions for working in domain of linear algebra, fourier transform, and matrices.

Why use NumPy ?

- In Python we have lists that serve the purpose of arrays, but they are slow to process.
- NumPy aims to provide an array object that is up to 50x faster than traditional Python lists.
- The array object in NumPy is called *ndarray*, it provides a lot of supporting functions.

Pandas is a Python library used for working with data sets.

- It has functions for analyzing, cleaning, exploring, and manipulating data.
- The name "Pandas" has a reference to both "Panel Data", and "Python Data Analysis"
- Pandas allows us to analyze big data and make conclusions based on statistical theories.
- Pandas can clean messy data sets, and make them readable and relevant.
- Pandas gives us answers about the data. Like:
 - Is there a correlation between two or more columns?
 - What is average value?
 - Max value?
 - Min value?
- Pandas are also able to delete rows that are not relevant, or contains wrong values, like empty or NULL values. This is called cleaning the data.

➤ DataFrame:

A Dataframe is a two-dimensional data structure, i.e., data is aligned in a tabular fashion in rows and columns. In dataframe datasets arrange in rows and columns, we can store any number of datasets in a dataframe. We can perform many operations on these datasets like arithmetic operation, columns/rows selection, columns/rows addition etc.

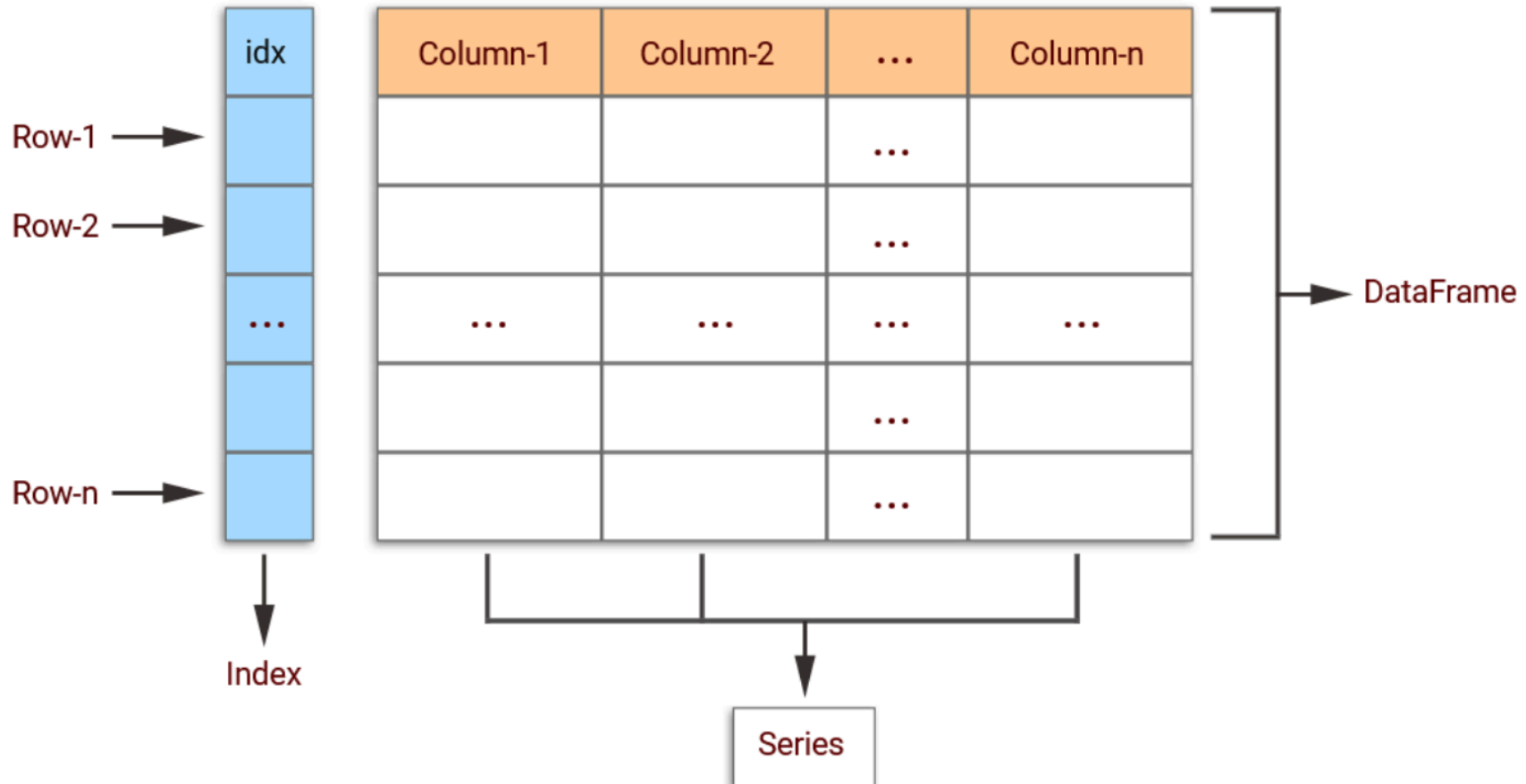
In the real world, a Pandas DataFrame will be created by loading the datasets from existing storage, storage can be SQL Database, CSV file, and Excel file. Pandas DataFrame can be created from the lists, dictionary, and from a list of dictionary etc.

In simple, a dataframe is a collection of Series and a Series is a collection of scalar values.

Pandas Dataframe



Pandas Data structure



- Importing Pandas
- Series Creation
- Data Frame Creation
- Reading Files using Pandas
- Writing to a File using Pandas
- Some functions offered by Pandas

`matplotlib` is a Python package used for data plotting and visualisation.

It is a useful complement to Pandas, and like Pandas, is a very feature-rich library which can produce a large variety of plots, charts, maps, and other visualisations.

Anatomy of a Matplotlib plot



1. **Import matplotlib** - For example, import `matplotlib.pyplot` as `plt`
2. **Prepare data** - This may be from an existing dataset (data analysis) or from the outputs of a machine learning model (data science)
3. **Setup the plot** - In other words, create the Figure and various Axes
4. **Plot data to the Axes** - Send the relevant data to the target Axes
5. **Customize the plot** - Add a title, decorate the colors, label each Axis
6. **Save and show** - See what your masterpiece looks like and save it to file if necessary

Matplotlib Workflow



```
# A matplotlib workflow
```

```
# 0. Import and get matplotlib ready
```

```
# %matplotlib inline # Not necessary in newer versions of Jupyter (e.g. 2022 onwards)
```

```
import matplotlib.pyplot as plt
```

```
# 1. Prepare data
```

```
x = [1, 2, 3, 4]
```

```
y = [11, 22, 33, 44]
```

```
# 2. Setup plot (Figure and Axes)
```

```
fig, ax = plt.subplots(figsize=(10,10))
```

```
# 3. Plot data
```

```
ax.plot(x, y)
```

```
# 4. Customize plot
```

```
ax.set(title="Sample Simple Plot", xlabel="x-axis", ylabel="y-axis")
```

```
# 5. Save & show
```

```
fig.savefig("../images/simple-plot.png")
```

Seaborn is a Python data visualization library based on matplotlib. It provides a high-level interface for drawing attractive and informative statistical graphics.

➤ <https://seaborn.pydata.org/>

Characteristics	Matplotlib	Seaborn
Use Cases	Matplotlib plots various graphs using Pandas and Numpy	Seaborn is the extended version of Matplotlib which uses Matplotlib along with Numpy and Pandas for plotting graphs
Complexity of Syntax	It uses comparatively complex and lengthy syntax.	It uses comparatively simple syntax which is easier to learn and understand.
Multiple figures	Matplotlib has multiple figures can be opened	Seaborn automates the creation of multiple figures which sometimes leads to out of memory issues
Flexibility	Matplotlib is highly customizable and powerful.	Seaborn avoids a ton of boilerplate by providing default themes which are commonly used.